MedFusion

Multimodal Feature Fusion for COVID-19 Chest X-Ray Classification

DINOv3 ViT-B/16
768-d CLS Token

Diffusion U-Net
down_block[2] + mid @ t=250

Fused: 2304-d → MLP Head

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PROBLEM

COVID-19 chest X-ray interpretation is time-consuming and prone to variability. Automated classification across COVID-19, Normal, Lung Opacity, and Viral Pneumonia can triage patients faster and reduce diagnostic burden.

APPROACH

Freeze both encoders. Extract DINOv3 CLS token (768-d) + diffusion U-Net features from down_block[2] and mid_block at t=250, average-pooled and projected to 768-d each. Concatenate → 2304-d → 3-layer MLP. Only 1.9M of 201M parameters are trained.

RESULTS

92.49%	0.93	0.96	0.98
Test Accuracy	Macro F1	COVID Precision	VP Recall

KEY FINDINGS

- Satellite-domain diffusion features transfer usefully to medical imaging without fine-tuning
- DINOv3 semantic + U-Net structural features are complementary, boosting chest X-ray classification
- Zero COVID cases misclassified as Viral Pneumonia — clinically meaningful separation